RESEARCH STATEMENT | Naman Gupta

My research goal is to make technology safer for *every* individual. I study how technology shapes and exposes people to violence disproportionately, how people seek assistance, and how interventions and research practices can reduce and prevent violence. Gender-based violence (GBV) is a major public health problem: about 1 in 3 women worldwide [1] and 1 in 2 transgender people in the US [2] experience physical, sexual, and psychological violence from an intimate partner. Technology perpetuates GBV on an interpersonal level (intimate partner violence, stalking, and AI-generated porn) and on a structural level (trafficking, genocide, environmental violence). These harms are connected through the same power dynamic. For example, coercion from an intimate partner underpins similar coercive and oppressive dynamic embedded in Governments' use of crime databases to ethnically cleanse Indigenous peoples, at different scales.

Studying harms and safety mechanisms require a deep understanding of availability of health resources in the community, survivors' individual and cultural needs, and providers' ability to plan for survivors' safety effectively. Thus, I combine ethnographic, participatory, and design methods with public health, security and privacy, law, and Indigenous and feminist scholarship. I design and analyze **equitable sociotechnical interventions such as education, tools, and research and design practices, helping communities foster healing and safety**. I make contributions across two threads:

1. **Interventions for GBV survivors and frontline providers.** I have designed interventions to develop competency among GBV survivors and frontline providers (health practitioners, legal justice workers, and policymakers). These interventions are currently deployed in practice through Madison Tech Clinic (MTC). I designed an educational intervention for providers towards early detection, mitigation, and prevention of violence [In-review CSCW 2027a]. Addressing survivor needs, I have designed tools and advocacy practices to help collect and preserve evidence for legal injunctions [CSCW 2025, CHI 2026, CCS 2025, In-review USENIX 2027]. These interventions build on measurement of the prevalence of GBV in college contexts [CHI-LBW 2024] and characterization of survivors' support-seeking strategies and challenges faced before they reach out to frontline GBV providers [USENIX 2024, NDSS-LBW 2024].
2. **Research Design Practice to resist and liberate from structural violence:** Equitable interventions require studying violence, which however itself, may reproduce more harm to marginalized and vulnerable communities. I have created safe and culturally-sensitive methods, data, and design practices. I do this by studying how historically, infrastructures (criminal databases, AMBER alerts, AI data centers) violate and genocide Indigenous communities, epistemology, and land [In-review CSCW 2027b] and how communities resist, refuse, reclaim, and build new infrastructure for care, sovereignty, and liberation.

Across both threads, I show that safety holds where technology design and decision-making power sit with the affected community, and fails where protective systems are built without them. In doing so, my research has been received awards at premier HCI venues such as ACM CHI and CSCW and selected as a finalist in applied research competition at CSAW. My research is supported by **DOJ grant**[1] and Morgridge Fellowship for Community-Engaged Scholarship. I **return research to communities** through talks, panels, and workshops at cross-disciplinary venues such as Wisconsin State Capitol, American Educational Research Association, Cook County Human Trafficking Task Force, *Gaawiin Geyaabi ("no more")*, Women Are Sacred, Reclaiming Our Love, Native Nations Nursing, Helpers, and Healers Summit.

Research Process. Tackling GBV is inherently complex. The experiences of harm are governed by dynamics of intimate relationships, gender, race, geography, and sovereignty: an outcome of structural racism, health inequity, and technical design. My research is inherently interdisciplinary and informed by expert collaborations and theories in domestic violence, trafficking, law, sexual violence, education, and Indigenous health. I begin with building **long-term reciprocal relationships with providers and organizations**. Such an ethnographic engagement is essential to reduce the reflexive distance through sensitivity. GBV survivors and providers, especially in rural areas and Tribal Nations, hold expert knowledge of violence yet are underrepresented in academia, and studying violence can itself reproduce harm. I develop a thorough understanding of communities' urgent and long-term needs and barriers that could inform potential interventions. To understand the diversity of lived experiences, my data collection involves a range of techniques ranging from interviews, surveys, online archives, and auto/ethnography. I analyze the data with care and sensitivity rooted in feminist, decolonial, and Indigenous epistemology. I use the "Upstream Story" [12] framework (Figure 1) to identify harms and interventions that could benefit the communities the most. This framework is especially important as much of the research on GBV focuses on examining harms and building downstream interventions that only satisfies communities' immediate needs. Instead, I **build interventions that meet communities' needs (downstream), reduce harm by recognizing coping strategies (midstream), and build capacity to prevent violence (upstream)** in the first place. I iterate, assess effectiveness, and plan for long-term sustainability with organizations. During my PhD studies, I have set up a community-engaged research program that has resulted in high-stakes impact on communities, but also comes with many challenges: distressing content, logistically complexity, and emotionally demanding fieldwork. I will mentor students to purse such research by creating a sensitive environment that addresses these challenges while balancing student well-being and academic rigor.

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1 Current Research

Thread 1: Building Interventions for Survivors and Frontline Providers

Interventions like tech clinics play a crucial role in restoring power and agency to survivors, however, their limited availability is insufficient to prevent violence at scale. Therefore, scale up and decentralize tech clinics, it is important to measure the prevalence of tech-facilitated GBV, coping strategies and challenges faced by survivors before they contact a provider. My studies ask three questions: how many survivors face violence, seek support and from who, what challenges do they face, what providers need in order to help, and how clinical practice be improved.

College campuses are of the most vulnerable environment for youth, who are among the most prevalent users of technology. In a survey of 776 US college students in a large midwest R1 university [CHI-LBW 2024], more than 70% experienced tech-facilitated GBV. 60% faced unwanted calls and messages, 51% non-consensual sharing of intimate images, videos; or messages, and less than 1% faced spyware from an adversary. Over half (53%) did not seek support; of those who did, 93% relied solely on informal providers (friends and family), and 6% solely on formal providers (GBV advocates in shelters). This glaring mismatch in support-seeking statistics show the crucial importance of studying and designing for informal providers. To understand *why* survivors do not seek support and what happens when they do, I interviewed nine survivors who face tech-facilitated GBV and sought support from their community [USENIX 2024]. The study showed that they are often disbelieved, paired with ineffective advice to mitigate and cope with violence. Providers use unsafe technological mediums to communicate, which is especially dangerous when an abuser isolates them and escalate violence. I also demonstrate how survivors and informal providers, despite not being technical experts, form a care ecosystem by seeking help, and collaborating to provide help to others. The study has contributed towards **foundational knowledge base on survivors' support-seeking strategies and challenges**. These findings provide evidence that my approach to utilizing "The Upstream Story" framework can be extremely effective to prevent violence. Most of the research in IPV has focused on research to strengthen and improve formal providers, that only a minority of survivors utilize. Most survivors utilize informal providers. My studies provides a pathway to **design interventions that support the most survivors, strengthen informal ecosystems, and prevent violence** in the society. As an example intervention for college students, I will work with the university health clinics to build mandatory training material around digital consent and sexual violence in campuses.

Providers face knowledge gaps in identifying and responding to fast-changing forms of tech-facilitated GBV. Therefore, I designed **trainings for more than 200 GBV providers**, including health practitioners, legal justice workers, and policy-makers, through national networks (National Network to End Domestic Violence, National Indigenous Women's Resource Center, Lac Courte Oreilles Tribal Nation) and state coalitions (EndAbuseWI, The Network, Mending the Sacred Hoop). In pre- and post-surveys with 68 providers, we found that trainings increased providers' confidence in providing support, safety planning, and making referrals through community collaboration. This training can be reliably used by technology experts in collaboration with GBV providers across the world. Through auto-ethnographic, I also examined how my own positionality shaped the training content, logistical coordination with GBV organizations, and learning outcomes among providers. I developed a **design framework of 24 guiding questions** that training designers can use to build trainings in high-stakes contexts with long-term sustainability in mind.

Survivors who need legal protection, such as a restraining order or a divorce settlement, must often prove tech abuse in court. I helped examine barriers faced by survivors and providers in Wisconsin [CSCW 2025] (**awarded honorable mention top 5%**) and design *Sherloc* [CHI 2026] (**awarded best paper top 1%**), a tool that helps tech clinic consultants prepare an evidence report that survivors can use to get legal injunctions. Similarly, *Lestrade* (**finalist at CSAW applied research competition**) [CCS 2025, In-review USENIX 2027], that helps consultants, advocates, and survivors investigate account compromise using data exports with less friction and uncertainty. I used my research and advocacy expertise to inform the design of the tool, which are currently being used into practice at MTC.

I introduced remote consultation service delivery model to MTC and directed its expansion in 26 urban and rural GBV organizations across 32 counties of Wisconsin. To date, the expansion has produced more than 50 remote consultations (25% of total consultations) with 15 rural organizations. I am now evaluating this hybrid service-delivery model and the role of advocates in clinical investigations, situating survivors' experiences of tech abuse in rural geographies and their service-seeking behavior. In the future, I want to establish a local, sovereign, and culturally-attuned tech clinic within each organizations to scale up the tech clinic service delivery.

Thread 2: Research Framework that Resist Structural Violence

This thread asks how infrastructure that claims to protect communities instead perpetuates violence among people, land, and knowledge systems, how communities resist it, towards a research framework that center relationality and sovereignty.

More than 84.3% of Indigenous women face violence when 97% perpetrators are non-native (settler) [13]. Indigenous women are 10x more likely to be murdered than the US national average [14] and represent 40% of sex trafficking victims across four sites in US and Canada [15]. Most tribal nations cannot prosecute settler perpetrators due to limits on jurisdictional sovereignty, giving rise to the Missing & Murdered Indigenous Relatives (MMIR) crisis, an ongoing genocide [16]. In a **first-ever study on MMIR from a sociotechnical perspective** [In-review CSCW 2027b], I built a measurement pipeline

to collect online archives that counter epistemic erasure: news articles, government reports, podcasts, and court hearings. I qualitatively analyzed 140 archives by embodying Indigenous epistemology through critical humility, relational accountability, and refusal. The archives show how modern infrastructure make Indigenous lives uncountable through bureaucracy instead of affording safety. For example, federal databases like NamUs only have 116 (or 2% of 5712) cases are recorded [17] and an Royal Canadian Mounted Police (RCMP) initiative to install 5G cell towers along Canada's "Highway of Tears" left out many stretches with spotty coverage. Most tribes also lack the access to criminal databases and the ability to release AMBER alerts. In response, Indigenous communities resist, refuse, and reclaim infrastructure as spaces of advocacy, ceremony, memory, and care: sovereign and crowd-sourced databases, digital maps for search and rescue, culturally grounded support for families, and social media advocacy. The paper contributes a **relational research framework that uses empirical methods to critically engage with structural oppression**. The research necessitates scholars to design for Indigenous sovereignty, jurisdictional tension, and governance of data infrastructure and platforms that must balance surveillance, privacy, and safety. University of Michigan Prof. Rebecca Irvine has invited me to collaborate on a book chapter on MMIR.

Ecological violence is tied to GBV in Indigenous communities. In North Dakota, the oil boom brought "Man camps" of transient that reportedly led to a 300% increase in sex crimes [18]. Hyperscale AI data centers now follow a similar pattern, sited in rural, drought-affected, and Black and Indigenous neighborhoods. I planned ethnographic interviews with data-center resistance organizers, who told me the urgency of their movement outweighed research participation. Respecting their refusal, I instead analyzed organization and government reports, news articles that communities themselves have made public. This ongoing work shows how extractive infrastructure severs relational ties to land and people, and creates conditions of exploitation: resource extraction, deepening health and economic inequities, worsening climate conditions, and loss of community agency over the future. Communities lead resistance movements that refuse DC and reclaim technologies to organize knowledge-sharing workshops, responsible land-use policies, and social media activism. This work reimagines how **sovereign and sustainable AI can benefit the communities** instead of accelerating oppression.

2 Future Agenda

Rapid scientific development of emerging technologies (e.g. wearables, alert systems, AI generated content) perpetuate violence among communities. In short-term, I will develop interventions that helps providers detect violence early or prevent it altogether. In parallel, I will research how technologies accelerate structural violence. And in long-term, I will theorize and build safe, ethical, responsible, and sovereign technologies that promote health equity that dismantle structural violence.

1. Capacity-building Interventions to Empower GBV Providers. This direction asks How can health and advocacy settings identify violence early without putting survivors at greater risk? Survivors typically complete intake forms at hospitals and shelters, which first responders such as emergency staff and sexual assault nurses use to assess psychological, economic, and physical safety. I will co-develop tech-facilitated GBV questions for these forms with providers, pair them with a layered referral system to expert support, and train first responders to recognize signs through short roll-call trainings. However, intake records can be seen by abusers, and screening can produce false positives and false negatives. I will treat these risks as a research problem: so the studies will measure safety harms and confidentiality alongside detection, and will build on existing healthcare screening instruments.

Another direction asks Can culturally grounded tech clinics and peer support extend digital safety to communities that current clinics do not reach? Only four tech clinics exist in the US (New York City, Madison, Seattle, and Arlington), so most survivors cannot reach one. Geographic isolation leads to access barriers such as limited internet and digital literacy, cultural stigma, close-knit communities, police siding with abusers, and slow justice systems. First, I will **(a) weigh survivors' cultural safety needs**. At MTC, we have conducted 50 remote consultations with 15 rural GBV organizations so far. Using this data, I will evaluate the effectiveness of such a service delivery model and how tech clinic consultants and rural advocates have navigated these consultations. Then, I will build a **(b) culturally attuned tech clinic** that responds to changing cultural safety needs, weaving the technology skills of youth with the teachings of Indigenous Elders. Second, to signal a shift towards collective healing, I will **(c) scale GBV providers to a peer support model** where healing happens with the community. I will **(d) co-design a peer-support clinic model** through a structured program that prepares providers or former survivors to run support groups where survivors share stories of violence and resilience [19]. These groups can redistribute epistemic and decision-making power, build technical skills, surface folk threat models, and move safety planning from individual plans toward collective safety.

2. Infrastructure, sovereignty, and health equity. This direction asks How do communities resist and rebuild oppressive infrastructure, and what does safe and sovereign technology look like? My work will continue to reckon with infrastructure that perpetuates structural violence faced by marginalized communities (e.g., war refugees, incarcerated, violence survivors in Global South). For example, what does privacy mean for exploited factory workers in India and Kenya who watch abusive content or wear cameras to train AI robots that will soon replace them? My projects will **(a) measure the privacy needs of communities subjected to structural violence** and how their lived experiences can inform interventions and tech policy. Importantly, to acknowledge their daily flourishing and resistance, I will build on my MMIR and AI Data Center work and **(b) examine how communities resist, refuse, and reclaim or build infrastructure for**

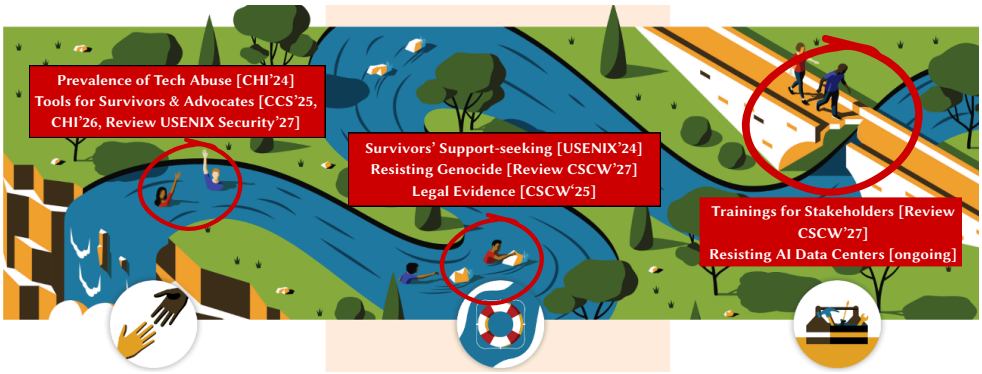


Figure 1: My work focuses on each part of the stream to effectively prevent violence by building interventions (a) downstream to support survivors' immediate needs after the violence has occurred, (b) midstream to support survivors' coping strategies to mitigate violence, and (c) upstream to prevent survivors from being exposed to the violence.

healing and justice. Recognizing culture as a vital safety condition, my work will **(c) design frameworks for cultural preservation and revitalization through technologies** through stories, knowledge, languages at-risk of extinction. Finally, collaborating with health epidemiologists and health practitioners, I will **(d) design technological visualizations tracing the systemic health inequities** with an aim towards providing culturally-competent interventions to curb them.

3. Diverse & Inclusive AI Safety Research. Finally, this direction asks Is the current AI safety research addressing harms effectively? Scholars have noted that computing research and interventions reflect an imperial impulse [20, 21]. However, without engaging with theories of change that center decoloniality, sovereignty, and liberation, safety research may result in siloed technical privacy fixes that fail to provide long-term safety. For example, trust & safety teams in Big Tech companies often tout their products as private, ethical, carbon-neutral, whereas, the same companies actively harm miners who extract rare earth minerals in Congo and assemble their products in India. My computing and critical theory expertise allows me to view technologies not just as a black box, but as a combination of complex social and technical components. My work contrasts how AI proliferation creates conditions of danger due to the fundamental value misalignment between the corporations and communities. I will conduct a **(a) review the value misalignment that underpin the digital safety research**. Then, I will expand "AI safety" broadly, to include harms across AI system's lifecycle, from the extraction of resources and labor to deployment, not only model behavior. Then, drawing on my workshops [In-review CHI 2027, In-review DIS 2026], I plan to **(b) create a diverse and inclusive framework for AI safety** that works for everyone. The framework will shape AI towards sustainability, sovereignty, and equity.

My Publications

[In-review CSCW 2027a] **Naman Gupta**, Connie W. Chau, Sophie Stephenson, Rahul Chatterjee, and Kate Walsh. "Lessons from Trauma-Informed Training on Technology-Facilitated Abuse for Gender-Based Violence Advocates". In: *review at CSCW* (2026). URL: https://arxiv.org/abs/2610.02688.

[CSCW 2025] Sophie Stephenson, **Naman Gupta**, Akhil Polamarasetty, Kyle Huang, David Youssef, Kayleigh Cowan, and Rahul Chatterjee. "Legal Evidence of Technology-Facilitated Abuse in Wisconsin: Surfacing Barriers Within and Beyond the Courtroom." In: *Computer Supported Cooperative Work and Social Computing* 🏆 **Honorable Mention (top 5%)**. 2025. URL: https://doi.org/10.1145/3757622.

[CHI 2026] Sophie Stephenson, **Naman Gupta**, Kyle Huang, David Youssef, Kayleigh Cowan, and Rahul Chatterjee. "Trauma-Informed Digital Evidence Collection:A Design Inquiry into Evidence Practices for Technology-Facilitated Abuse in Intimate Partner Violence". In: *2026 CHI Conference on Human Factors in Computing Systems* 🏆 **Best Paper (top 1%)**. 2026. URL: https://doi.org/10.1145/3772318.3790296.

[CCS 2025] Julia Nonnenkamp, **Naman Gupta**, Abhimanyu Dev Gupta, and Rahul Chatterjee. "Hidden in Plain Bytes: Investigating Interpersonal Account Compromise with Data Exports". In: *2025 ACM SIGSAC Conference on Computer and Communications Security*. 2025. URL: https://doi.org/10.1145/3719027.3765147.

[In-review USENIX 2027] Julia Nonnenkamp, **Naman Gupta**, Angela Wang, Sophie Stephenson, and Rahul Chatterjee. "LEStrADE: Investigating Account Compromise using Data Exports for Survivors of Technology-Facilitated Abuse". In: *review at USENIX Security*. 2026. URL: https://naman.github.io/assets/files/papers/lestrade.pdf.

[CHI-LBW 2024] **Naman Gupta**, Sanchari Das, Kate Walsh, and Rahul Chatterjee. "A Critical Analysis of the Prevalence of Technology-Facilitated Abuse in US College Students". In: *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems*. 2024. URL: https://doi.org/10.1145/3613905.3652036.

[USENIX 2024] **Naman Gupta**, Kate Walsh, Sanchari Das, and Rahul Chatterjee. ""I really just leaned on my community for support": Barriers, Challenges, and Coping Mechanisms Used by Survivors of {Technology-Facilitated} Abuse to Seek Social Support". In: 2024. URL: https://dl.acm.org/doi/10.5555/3698900.3699179.

[NDSS-LBW 2024] **Naman Gupta**, Rose Ceccio, Majed Almansoori, and Rahul Chatterjee. "Analyzing the Patterns and Behavior of Users When Detecting and Preventing Tech-enabled Stalking". In: 2023. URL: https://doi.org/10/gr8rnx.

[In-review CSCW 2027b] **Naman Gupta**, Sophie Stephenson, Chung Chi Yeung, Wei Ting Wu, Jeneile Luebke, Kate Walsh, and Rahul Chatterjee. ""Lighting The Way For Those Not Here": How Technology Researchers Can Help Fight the Missing and Murdered Indigenous Relatives (MMIR) Crisis". In: *review at CSCW 2027*. 2025. URL: https://doi.org/10.48550/arXiv.2601.17966.

[In-review CHI 2027] **Naman Gupta**, Freddy Reiber, Jasmine C. Foriest, Ailsa Robertson, Jordan Taylor, Palak Dudani, Ghadeer Awwad, Shreya Chowdhary, David Gamba, Eugene Kim, and Katie Seaborn. "Counter-Computing: Digital Resistances and Global Solidarities". In: *Review at ACM CHI Conference on Human Factors in Computing Systems*. 2027.

[In-review DIS 2026] Jasmine C Foriest, Alicia DeVrio, Li Qiwei, Nimra Ahmed, **Naman Gupta**, Jensine Raihan, Franchesca Spektor, and Justine Zhang. "Violence as a Boundary Object: Language, Power, and Design Justice". In: *Review at ACM DIS Designing Interactive Systems*. 2026.

References

[1] World Health Organization. *Violence against women prevalence estimates, 2023: global overview*. en. Aug. 2026.

[2] Maya Novick, Allyson Blackburn, and Rachel C. Garthe. "Intimate Partner Violence Against Transgender and Gender Expansive Individuals: A Review of Existing Literature". In: *Journal of Undergraduate Social Work Research* (2023).

[12] National Academies of Sciences, Engineering, and Medicine, Committee on Integrating Social Needs Care into the Delivery of Health Care to Improve the Nation's Health, Board on Health Care Services, and Health and Medicine Division. *Integrating Social Care into the Delivery of Health Care: Moving Upstream to Improve the Nation's Health*. The National Academies Press. The National Academies Press, 2019.

[13] Andrè B Rosay. "Violence Against American Indian and Alaska Native Women and Men". In: *National Institute of Justice* (2016).

[14] Ronet Bachman, Heather Zaykowski, Rachel Kallmyer, Margarita Poteyeva, and Christina Lanier. "Violence Against American Indian and Alaska Native Women and the Criminal Justice Response: What is Known". In: (2008).

[15] National Congress of American Indians Policy Research Center. *Human & Sex Trafficking: Trends and Responses Across Indian Country*. Tech. rep. 2016.

[16] National Inquiry into Missing and Murdered Indigenous Women and Girls. *Reclaming power and place. The final report of the national inquiry into missing and murdered indigenous women and girls.* Tech. rep. June 2019. URL: https://www.mmiwg-ffada.ca/wp-content/uploads/2019/06/Final_Report_Vol_1a-1.pdf (visited on 04/04/2024).

[17] Annita Lucchesi and Abigail Echo-Hawk. *Missing and Murdered Indigenous Women & Girls: A snapshot of data from 71 urban cities in the United States*. Tech. rep. 2018.

[18] Kelly Yu. *Indigenous Women, Fracking, and Violence*. en. June 2013. URL: https://womensearthalliance.org/wea-voices/indigenous-women-fracking-and-violence/ (visited on 05/08/2025).

[19] Max Osborn, Tanyanne Ball, and Valli Rajah. "Peer Support Work in the Context of Intimate Partner Violence: A Scoping Review". In: *Trauma, Violence, & Abuse* (2024).

[20] Paul Dourish and Scott D. Mainwaring. "Ubicomp's colonial impulse". In: *Proceedings of the 2012 ACM Conference on Ubiquitous Computing*. UbiComp '12. Association for Computing Machinery, 2012.

[21] Jensine Raihan. "The Colonial Legacy of the Human in Human-Computer Interaction: Implications for Critical Design Approaches". In: *Proc. Sixth Decenn. Aarhus Conf. Comput. X Crisis*. AAR '25. Association for Computing Machinery, 2025.